

ML Engineering for Production (DSAI 406)

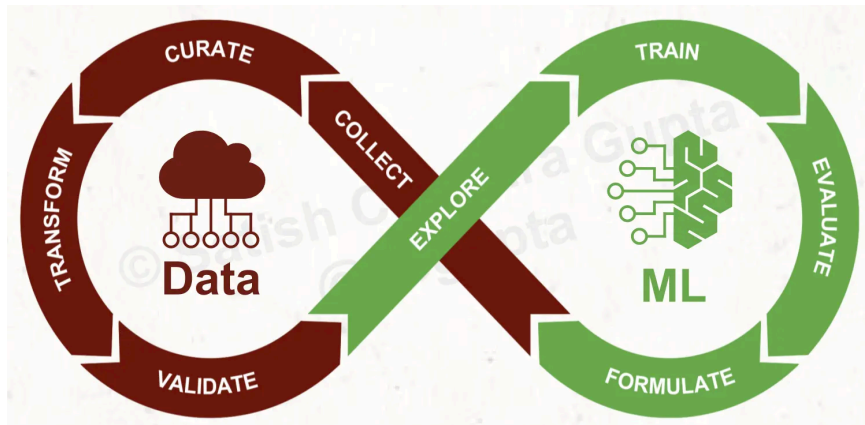
Lecture 1

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Recording is NOT allowed

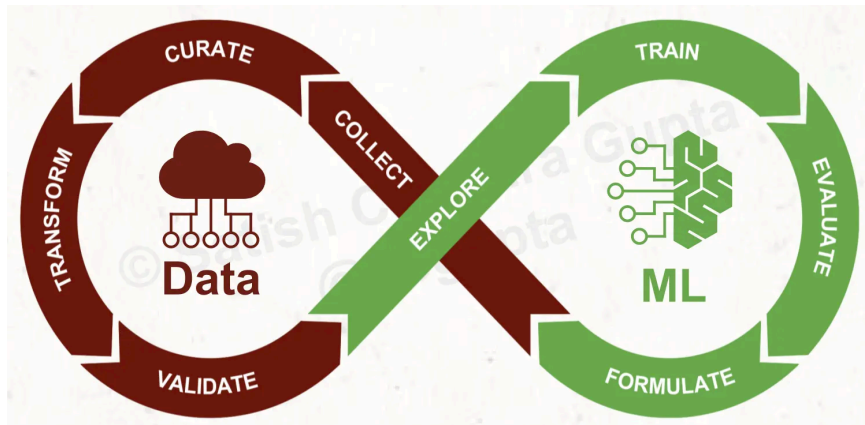
Model Development

- Model Development: *Data (CSV)* → *Notebook* → *Feature Engineering* → *Model training* → *Accuracy*



Model Development

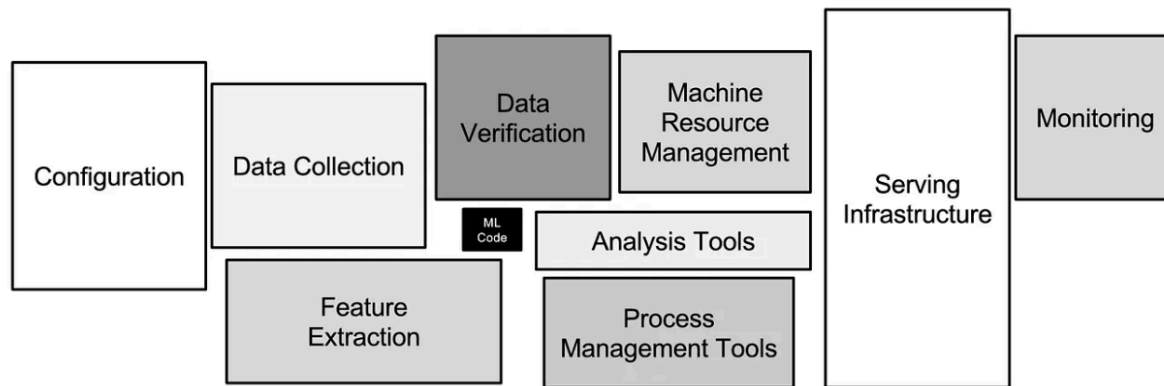
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- Imagine a scenario with a small change in library version when deploying it into production (Reproducibility)

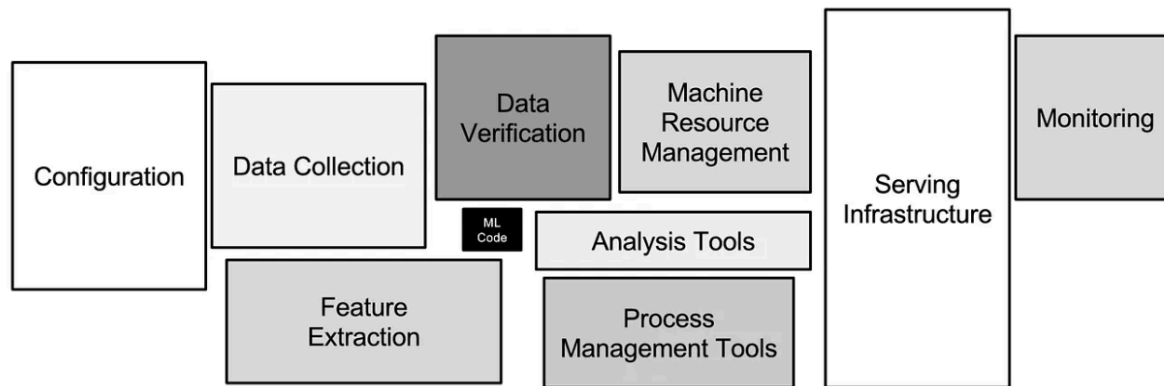
Hidden Technical Debt in Machine Learning Systems, NeurIPs (2015)

ML systems require robust engineering beyond just algorithm development



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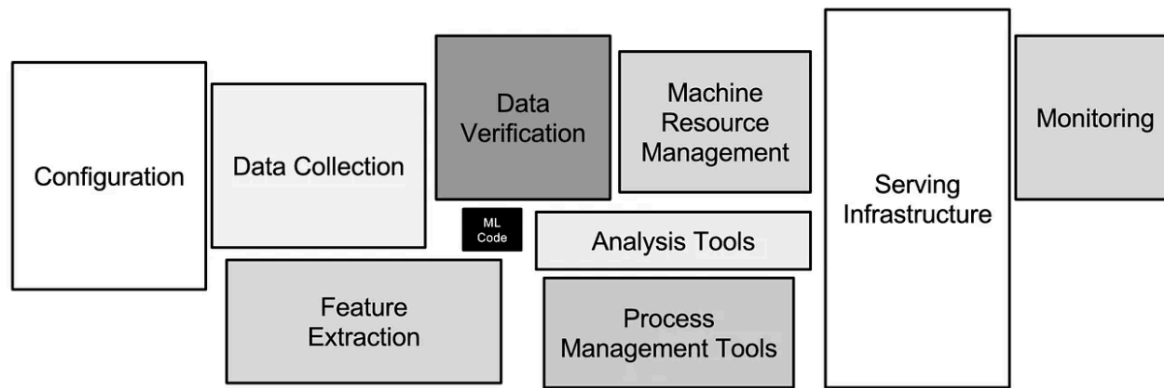
ML systems require robust engineering beyond just algorithm development



The goal is not to add new functionality, but to enable future improvements, reduce errors, and improve maintainability.

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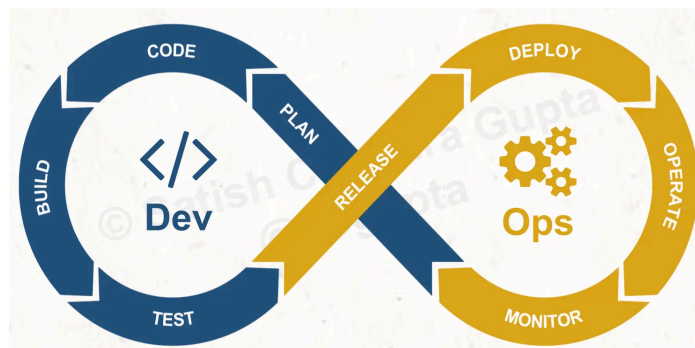


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MLOps is about **Automation + Shipping**

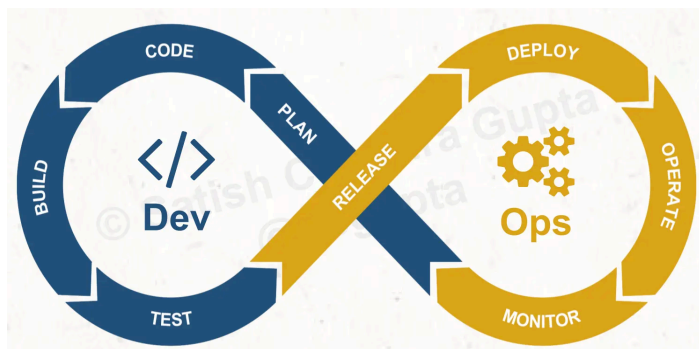
What is DevOps?

- Increase an organization's velocity in releasing high-quality software
- Speed, reliability, scale, security, continuous integration and delivery, and microservices



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Tasks:

- Versioning Git (Code)
- Unit and integration tests

What is DataOps?

- Periodic collection of data
- Streaming data
- Event-driven data
- Big data jobs
- Data versioning

What is DataOps?

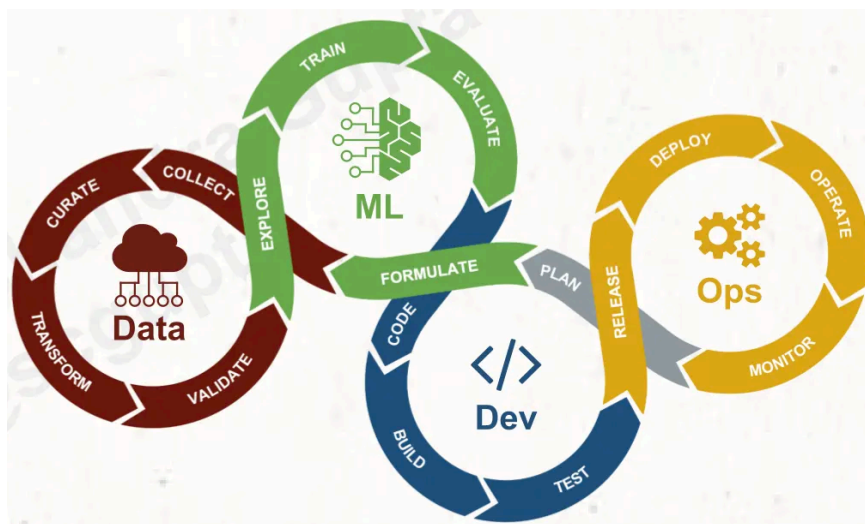
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Tasks:

- Versioning DVC
- Data Validation

What is MLOps?

- MLOps is the intersection of Machine Learning, DevOps, and Data Engineering
- MLOps as the process of automating machine learning using DevOps methodologies



What is MLOps?

Additional monitoring and instrumentation must account for new things that can break, like data drift

Tasks:

- Versioning Git (Code) + DVC (Data) + Model Registry
- Unit and integration tests + Data Validation and Model Quality
- Deploying + Deploying a prediction pipeline instead of just a binary/service
- Models decay (Data Drift)

What are MLOps Topics?

- Continuous improvement / integration / delivery
- Cloud Computing
- AutoML
- Containers
- Edge Computing
- Model Portability

What is the difference between data scientist and ML Engineer and ML Researcher ?

References

- Images were taken from ML4devs
- Research paper "Hidden Technical Debt in Machine Learning Systems, NeurIPS (2015) by Google

Learn More

Course Homepage

Lessons Learned

- All assignments should be uploaded to classroom even if it was graded
- All assignments should be uploaded before the deadline
- Medical excuses should be submitted within one week of the excused task (assignment, quiz, midterm)
- Contents of the redo tasks will be based on the content up to the date of the task remake not the original content